

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

DETERMINANTS

DIBY (DO IT BY YOURSELF)

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DIBY-01

The cofactor of the element '4' in the determinant $\begin{vmatrix} 1 & 3 & 5 & 1 \\ 2 & 3 & 4 & 2 \\ 8 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \end{vmatrix}$ is

- (A) 4 (B) 10 (C) -10 (D) -4

[Ans.: (B)]

DIBY-02

If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, then D can not be (where A_i, B_i and C_i $i = 1, 2, 3$)

- (A) $c_1C_1 + c_2C_2 + c_3C_3$ (B) $a_1A_1 + a_2A_2 + a_3A_3$
 (C) $a_2A_1 + b_2B_1 + c_2C_1$ (D) $a_1A_1 + b_1B_1 + c_1C_1$

[Ans.: (c)]

DIBY-03

If $A = \begin{vmatrix} 5 & 6 & 3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{vmatrix}$, then cofactors of the elements of second row are

- (A) 39, -3, 11 (B) -39, 3, 11 (C) -39, 27, 11 (D) -39, -3, 11

[Ans. : (C)]

DIBY-04

The solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$, is

[IIT-JEE-1981]

[Ans. : $x = -1, 2$]

DIBY-05

If $A = \begin{vmatrix} 3 & 2 & 4 \\ 1 & 2 & 1 \\ 3 & 2 & 6 \end{vmatrix}$ and A_{ij} are the cofactors of a_{ij} , then $a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ is equal to

- (A) 8 (B) 6 (C) 4 (D) 0

[Ans. : (A)]

DIBY-06

Consider the determinant $A = \begin{vmatrix} 2 & -1 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 3 & 1 & -1 & 1 \\ 0 & -1 & 5 & 0 \end{vmatrix}$. Value of A is

(A) 45

(B) -45

(C) 9

(D) -9

[Ans. : (B)]

DIBY-07

The number of real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$ is:

(A) 0

(B) 1

(C) 2

(D) 3

[Ans. : (D)]

DIBY-08

If $\begin{vmatrix} b+c & c & b \\ c & c+a & a \\ b & a & a+b \end{vmatrix} = k(abc)$, then value of k is _____.

[Ans. : 4]

DIBY-09

Show that $\begin{vmatrix} b+c & a+b & a \\ c+a & b+c & b \\ a+b & c+a & c \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$.

DIBY-10

Prove that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$.

DIBY-11

If $\begin{vmatrix} a^2 & bc & ac + c^2 \\ a^2 + ab & b^2 & ac \\ ab & b^2 + bc & c^2 \end{vmatrix} = ma^n b^n c^n$, then

- (A) $m + n = 6$ (B) $m + n = 4$ (C) $m - n = 0$ (D) $m - n = 2$

[Ans. : A, D]

DIBY-12

If $A = \begin{vmatrix} 5 & 6 & 12 \\ 3 & 1 & 5 \\ 5 & 3 & 7 \end{vmatrix} + \begin{vmatrix} 12 & 6 & 5 \\ 5 & 1 & 3 \\ 7 & 3 & 5 \end{vmatrix}$, $B = \begin{vmatrix} 2024 & 2025 & 2026 \\ 2027 & 2028 & 2029 \\ 2020 & 2021 & 2022 \end{vmatrix} + \begin{vmatrix} 2024 & 2025 & 2026 \\ 2020 & 2021 & 2022 \\ 2027 & 2028 & 2029 \end{vmatrix}$

the value of $\sec^2 A + \sec^2 B - \sin^2 A$ is

- (A) 1 (B) 2 (C) 0 (D) 3

[Ans. : B]

DIBY-13

The value of determinant $\begin{vmatrix} a & a+b & a+2b \\ a+2b & a & a+b \\ a+b & a+2b & a \end{vmatrix}$ is

(A) $9a^2(a+b)$

(B) $9b^2(a+b)$

(C) $3b^2(a+b)$

(D) $7a^2(a+b)$

[Ans. : B]

DIBY-14

The value of determinant $\begin{vmatrix} a & a+b & a+2b \\ a+2b & a & a+b \\ a+b & a+2b & a \end{vmatrix}$ is

(A) $9a^2(a+b)$

(B) $9b^2(a+b)$

(C) $3b^2(a+b)$

(D) $7a^2(a+b)$

[Ans. : B]

DIBY-15

If $D = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ and $D' = \begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix}$, then prove that $D' = 2D$.

DIBY-16

For positive numbers x, y and z , the numerical value of the determinant

$$\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix} \text{ is } \underline{\hspace{2cm}}.$$

DIBY-17

Let a, b, c be such that $b(a + c) \neq 0$.

$$\text{If } \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$$

then the value of n is

- | | |
|----------------------|-----------------|
| (A) Any even integer | (B) Any integer |
| (C) Any odd integer | (D) Zero |

[AIEEE-2009]

DIBY-18

If $a + x = b + y = c + z + 1$, where a, b, c, x, y, z are non-zero distinct real numbers, then

$$\begin{vmatrix} x & a + y & x + a \\ y & b + y & y + b \\ z & c + y & z + c \end{vmatrix} \text{ is equal to}$$

[JEE (Main)-2020]

- (A) $y(b - a)$ (B) $y(a - b)$ (C) $y(a - c)$ (D) 0

DIBY-19

Show that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a - b)(b - c)(c - a)(a^2 + b^2 + c^2 + ab + bc + ca)$

DIBY-20

The value of determinant $\begin{vmatrix} ka & k^2 + a^2 & 1 \\ kb & k^2 + b^2 & 1 \\ kc & k^2 + c^2 & 1 \end{vmatrix}$ is

- (A) $k^2(a-b)(b-c)(c-a)$ (B) $kabc(a-b)(b-c)(c-a)$
 (C) $k(a-b)(b-c)(c-a)$ (D) $k(a+b-c)(b+c-a)(c+a-b)$

[Ans. : (C)]

DIBY-21

Let $\Delta_1 = \begin{vmatrix} ap^2 & 2ap & 1 \\ aq^2 & 2aq & 1 \\ ar^2 & 2ar & 1 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} apq & a(p+q) & 1 \\ aqr & a(q+r) & 1 \\ arp & a(r+p) & 1 \end{vmatrix}$, then

[JEE Mains-2020]

- (A) $\Delta_1 = \Delta_2$ (B) $\Delta_2 = 2\Delta_1$ (C) $\Delta_1 = 2\Delta_2$ (D) $\Delta_1 + \Delta_2 = 0$

[Ans. : (B)]

DIBY-22

If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)(x+a+b+c)^2, x \neq 0$ and $a+b+c \neq 0$, then x

is equal to

[JEE (Main)-2019]

- (A) $2(a+b+c)$ (B) $-(a+b+c)$ (C) abc (D) $-2(a+b+c)$

[Ans. : (D)]

DIBY-23

Without expanding prove that: $\begin{vmatrix} 23 & 66 & 11 \\ 36 & 55 & 26 \\ 63 & 143 & 37 \end{vmatrix} = 0$

DIBY-24

Solve for λ : $\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix} = 0$

[Ans. : $(\lambda = 0, -(a^2 + b^2 + c^2))$]

DIBY-25

In triangle ABC , if
$$\begin{vmatrix} 1 & 1 & 1 \\ \cot \frac{A}{2} & \cot \frac{B}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} + \tan \frac{C}{2} & \tan \frac{C}{2} + \tan \frac{A}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \end{vmatrix} = 0$$
, then the triangle must be

(A) Equilateral (B) Isosceles (C) Obtuse angled (D) None of these

[Ans. : (B)]

DIBY-26

Prove that
$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

DIBY-27

Prove that
$$\begin{vmatrix} x^2 & x^2 - (y-z)^2 & yz \\ y^2 & y^2 - (z-x)^2 & zx \\ z^2 & z^2 - (x-y)^2 & xy \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)(x^2 + y^2 + z^2).$$

DIBY-28

Show that
$$\begin{vmatrix} a-b+c & a+b-c & a-b-c \\ b-c+a & b+c-a & b-c-a \\ c-a+b & c+a-b & c-a-b \end{vmatrix} = 4(a^3 + b^3 + c^3 - 3abc).$$

DIBY-29

The values of α , for which
$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0,$$
 lie in the interval. **[JEE Mains-2024]**

(A) (2, 1)

(B) (-3, 0)

(C) $\left(-\frac{3}{2}, \frac{3}{2}\right)$

(D) (0, 3)

[Ans. : (B)]

DIBY-30

For $\alpha, \beta \in R$ and a natural number n , let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2} + \alpha \\ 2r & 2 & n^2 - \beta \\ 3r - 2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10} - A_8$ is

[JEE Mains-2024]

- (A) $4\alpha + 2\beta$ (B) $2\alpha + 4\beta$ (C) $2n$ (D) 0

[Ans. : (A)]

DIBY-31

If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x + 81)$, then $\lambda, \frac{\lambda}{3}$ are the roots of the equation

[JEE Mains-2023]

- (A) $4x^2 + 24x - 27 = 0$ (B) $4x^2 - 24x + 27 = 0$
 (C) $4x^2 + 24x + 27 = 0$ (D) $4x^2 - 24x - 27 = 0$

[Ans. : (B)]

DIBY-32

Let m and M be respectively the minimum and maximum values of

$$\begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 + \cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1 + \sin 2x \end{vmatrix}. \text{ Then the ordered pair } (m, M) \text{ is equal to:}$$

[JEE Mains-2020]

- (A) $(1, 3)$ (B) $(-3, -1)$ (C) $(-4, -1)$ (D) $(-3, 3)$

[Ans. : (B)]

DIBY-33

The maximum value of $f(x) = \begin{vmatrix} \sin^2 x & 1 + \cos^2 x & \cos 2x \\ 1 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}, x \in R$

[JEE Mains-2021]

- (A) $\frac{3}{4}$ (B) 5 (C) $\sqrt{5}$ (D) $\sqrt{7}$

[Ans. : (C)]

DIBY-34

Let $f(x) = \begin{vmatrix} \sin^2 x & -2 + \cos^2 x & \cos 2x \\ 2 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1 + \cos 2x \end{vmatrix}$, $x \in [0, \pi]$. Then the maximum value of $f(x)$ is equal to

[JEE Mains-2021]

[Ans. : 6]

DIBY-35

Value of determinant $\begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix}$ is

- (A) $4a^2b^2c^2$ (B) $3abc$ (C) $1 + a^2 + b^2 + c^2$ (D) $2 - a^2 + b^2 + c^2$

[Ans. : (C)]

DIBY-36

The determinant $\begin{vmatrix} xp + y & x & y \\ yp + z & y & z \\ 0 & xp + y & yp + z \end{vmatrix} = 0$, If

[IIT-JEE-1997]

- (A) x, y, z are in A.P. (B) x, y, z are in G.P.
(C) x, y, z are in H.P. (D) xy, yz, zx are in A.P.

[Ans. : (B)]

DIBY-37

If a, b are unequal real numbers, then $\begin{vmatrix} 1 & 1 & 1 \\ (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (x-b)(x-c) & (x-c)(x-a) & (x-a)(x-b) \end{vmatrix} = 0$

when

- (A) $a + b + c = 0$ (B) $x = \frac{1}{3}(a + b + c)$
(C) $x = \frac{1}{2}(a + b + c)$ (D) $x = a + b + c$

[Ans. : (B)]

DIBY-38

Let the three digit numbers $A28, 3B9$ and $62C$, where A, B and C are integers between 0 and 9, be divisible by a fixed integer k .

Show that the determinant $\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix}$ is divisible by k .

[IIT-JEE-1990]

DIBY-39



Prove that in a $\triangle ABC$, $\begin{vmatrix} \sin 2A & \sin C & \sin B \\ \sin C & \sin 2B & \sin A \\ \sin B & \sin A & \sin 2C \end{vmatrix} = 0$.

DIBY-40

Show: $\begin{vmatrix} (a_1 - b_1)^2 & (a_1 - b_2)^2 & (a_1 - b_3)^2 \\ (a_2 - b_1)^2 & (a_2 - b_2)^2 & (a_2 - b_3)^2 \\ (a_3 - b_1)^2 & (a_3 - b_2)^2 & (a_3 - b_3)^2 \end{vmatrix} = \begin{vmatrix} (1 + a_1 b_1)^2 & (1 + a_1 b_2)^2 & (1 + a_1 b_3)^2 \\ (1 + a_2 b_1)^2 & (1 + a_2 b_2)^2 & (1 + a_2 b_3)^2 \\ (1 + a_3 b_1)^2 & (1 + a_3 b_2)^2 & (1 + a_3 b_3)^2 \end{vmatrix}$

DIBY-41



Show $\Delta = \begin{vmatrix} \sin(x+A) & \cos(x+A) & a+x\sin A \\ \sin(x+B) & \cos(x+B) & b+x\sin B \\ \sin(x+C) & \cos(x+C) & c+x\sin C \end{vmatrix}$ is independent of 'x'.

DIBY-42

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}$, $x \in \mathbb{R}$, then $\frac{d^2 y}{dx^2} + y$ is equal to [JEE Mains-2025]

(A) -1 (B) 28 (C) 27 (D) 1

[Ans.: (A)]

CRAMER'S RULE



DIBY-43

If the system of equations

$$2x + y - z = 5$$

$$2x - 5y + \lambda z = \mu$$

$$x + 2y - 5z = 7$$

has infinitely many solutions, then $(\lambda + \mu)^2 + (\lambda - \mu)^2$ is equal to

- (A) 916 (B) 912 (C) 920 (D) 904

[Ans.: (A)]

DIBY-44

For the system of linear equations

$$2x + 4y + 2az = b, x + 2y + 3z = 4, 2x - 5y + 2z = 8$$

which of the following is NOT correct?

- (A) It has infinitely many solutions if $a = 3, b = 6$
(B) It has unique solution if $a = b = 6$
(C) It has unique solution if $a = b = 8$
(D) It has infinitely many solution if $a = 3, b = 8$

[Ans.: (A)]

[JEE Mains-2023]

[JEE Mains-2023]

CRAMER'S RULE



DIBY-45

If the system of linear equations

$$7x + 11y + \alpha z = 13, 5x + 4y + 7z = \beta, 175x + 194y + 57z = 361$$

has infinitely many solutions, then $\alpha + \beta + 2$ is equal to

(A) 4

(B) 3

(C) 5

(D) 6

[Ans. : (A)]

DIBY-46

For $\alpha, \beta \in R$, suppose the system of linear equations

$$x - y + z = 5, 2x + 2y + \alpha z = 8, 3x - y + 4z = \beta$$

has infinitely many solutions. Then α and β are the roots of

(A) $x^2 - 10x + 16 = 0$

(B) $x^2 + 18x + 56 = 0$

(C) $x^2 - 18x + 56 = 0$

(D) $x^2 + 14x + 24 = 0$

[Ans. : (C)]

[JEE Mains-2023]

[JEE Mains-2023]

CRAMER'S RULE



DIBY-47

If the system of linear equations

$$2x + 3y - z = -2, x + y + z = 4, x - y + |\lambda|z = 4\lambda - 4$$

where $\lambda \in \mathbb{R}$, has no solution, then

- (A) $\lambda = 7$ (B) $\lambda = -7$ (C) $\lambda = 8$ (D) $\lambda^2 = 1$

[Ans. : (B)]

DIBY-48

The system of equations

$$-kx + 3y - 14z = 25, -15x + 4y - kz = 3, -4x + y + 3z = 4$$
 is consistent for all k in the set

[JEE Mains-2022]

- (A) R (B) $R - \{-11, 13\}$ (C) $R - \{13\}$ (D) $R - \{-11, 11\}$

[Ans. : (D)]

CRAMER'S RULE



DIBY-49

The number of values of α for which the system of equations:

[JEE Mains-2022]

$x + y + z = \alpha, \alpha x + 2\alpha y + 3z = -1, x + 3\alpha y + 5z = 4$ is inconsistent, is

- (A) 0 (B) 1 (C) 2 (D) 3

[Ans.: (B)]

DIBY-50

If the system of linear equations

$2x + y - z = 7, x - 3y + 2z = 1, x + 4y + \delta z = k$, where $3, k \in R$

has infinitely many solutions, then $\delta + k$ is equal to :

[JEE Mains-2022]

- (A) -3 (B) 3 (C) 6 (D) 9

[Ans.: (B)]

DIBY-51

If the system of linear equations $2x - 3y = \gamma + 5, \alpha x + 5y = \beta + 1$, where $\alpha, \beta, \gamma \in R$ has

infinitely many solutions, then the value of $|9\alpha + 3\beta + 5\gamma|$ is equal to [JEE Mains-2022]

[Ans.: 58]

CRAMER'S RULE



DIBY-52

The greatest value of $c \in R$ for which the system of linear equations

$x - cy - cz = 0, cx - y + cz = 0, cx + cy - z = 0$ has a non-trivial solution, is:

[JEE Mains-2019]

- (A) -1 (B) $1/2$ (C) 2 (D) 0

[Ans.: (B)]

DIBY-53

The set of all values of λ for which the system of linear equations has non-trivial

solution $x - 2y - 2z = \lambda x, x + 2y + z = \lambda y, -x - y = \lambda z$

[JEE Mains-2019]

- (A) is a singleton (B) contains exactly two elements
(C) is an empty set (D) contains more than two elements

[Ans.: (A)]

DIBY-54

The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ has a non-trivial solution, is

[JEE Mains-2019]

(A) 3

(B) 2

(C) 4

(D) 1

[Ans.: (B)]

DIBY-55

If the system of equations $2x - y + z = 4, 5x + \lambda y + 3z = 12, 100x - 47y + \mu z = 212$ has infinitely many solutions, then $\mu - 2\lambda$ is equal to

[JEE Mains-2025]

(A) 57

(B) 59

(C) 55

(D) 56

[Ans.: (A)]

CRAMER'S RULE



DIBY-56

If the system of equations:

$$2x + 3y - z = 5$$

$$x + \alpha y + 3z = -4$$

$3x - y + \beta z = 7$ has infinitely many solutions, then $13\alpha\beta =$

(A) 1110

(B) 1120

(C) 1210

(D) 1220

[JEE Mains-2024]

[Ans. : (B)]

DIBY-57

For $\alpha, \beta \in \mathbb{R}$, suppose the system of linear equations

$$x - y + z = 5$$

$$2x + 2y + \alpha z = 8$$

$3x - y + 4z = \beta$ has infinitely many solutions. Then α and β are the roots of

[JEE Mains-2023]

(A) $x^2 - 10x + 16 = 0$

(B) $x^2 + 18x + 56 = 0$

(C) $x^2 - 18x + 56 = 0$

(D) $x^2 - 14x + 24 = 0$

[Ans. : (C)]

DIBY-58

If the system of equations

$$(\lambda - 1)x + (\lambda - 4)y + \lambda z = 5,$$

$$\lambda x + (\lambda - 1)y + (\lambda - 4)z = 7,$$

$(\lambda + 1)x + (\lambda + 2)y - (\lambda + 2)z = 9$ has infinitely many solutions, then $\lambda^2 + \lambda$ is equal to:

[JEE Mains-2025]

(A) 6

(B) 10

(C) 20

(D) 12

[Ans. : (D)]

DIBY-59

Let the system of equations $x + 5y - z = 1$; $4x + 3y - 3z = 7$; $24x + y + \lambda z = \mu$; $\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system, if x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is

[JEE Mains-2025]

(A) 3

(B) 6

(C) 5

(D) 4

[Ans. : (A)]

DIBY- 60

If the equations $x + y = 1$, $(c + 2)x + (c + 4)y = 6$, $(c + 2)^2x + (c + 4)^2y = 36$ are consistent, then find c .

[Ans: $c = 2, 4$]

DIBY- 61

Let the system of linear equations

$$x + y + \alpha z = 2; 3x + y + z = 4; x + 2z = 1$$

have a unique solution (x^*, y^*, z^*) . If (α, x^*) , (y^*, α) and $(x^*, -y^*)$ are collinear points, then the sum of absolute values of all possible values of α is

[JEE Mains-2022]

(A) 4

(B) 3

(C) 2

(D) 1

[Ans: C]

DIBY- 62 (a)

If the trivial solution is the only solution of the system of equations

$$x - ky + z = 0, kx + 3y - kz = 0, 3x + y - z = 0$$

then the set of all values of k is

[AIEEE-2011]

- (A) $R - \{-3\}$ (B) $\{2, -3\}$ (C) $R - \{2, -3\}$ (D) $R - \{2\}$

[Ans: C]

DIBY- 62 (b)

Let S be the set of all integer solutions, (x, y, z) , of the system of equations

$$x - 2y + 5z = 0, -2x + 4y + z = 0, -7x + 14y + 9z = 0$$

such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to

[JEE Mains - 2020]

[Ans: 08]

DIBY- 63

If the system of linear equations

$$x + ky + 3z = 0, 3x + ky - 2z = 0, 2x + 4y - 3z = 0$$

has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to

- (A) -10 (B) 10 (C) -30 (D) 30

[JEE (Main)-2018]

[Ans: B]

DIBY- 64

The system of linear equations

$$x + y + z = 2, 2x + 3y + 2z = 5, 2x + 3y + (a^2 - 1)z = a + 1$$

- (A) has infinitely many solutions for $a = 4$
 (B) is inconsistent when $|a| = \sqrt{3}$
 (C) has a unique solution for $|a| = \sqrt{3}$
 (D) is inconsistent when $a = 4$

[JEE (Main)-2019]

[Ans: B]

DIBY- 65



If the system of linear equations

$$x - 4y + 7z = g, 3y - 5z = h, -2x + 5y - 9z = k$$

is consistent, then

- (A) $g + h + k = 0$ (B) $g + 2h + k = 0$ (C) $g + h + 2k = 0$ (D) $2g + h + k = 0$

[JEE (Main)-2019]

[Ans: D]

DIBY- 66

If $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$; then

for all $\theta \in \left(0, \frac{\pi}{2}\right)$

- (A) $\Delta_1 + \Delta_2 = -2x^3$ (B) $\Delta_1 - \Delta_2 = -2x^3$
(C) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$ (D) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$

[JEE (Main)-2019]

[Ans: A]

For the system of linear equations : $x - 2y = 1$, $x - y + kz = -2$, $ky + 4z = 6$, $k \in R$, consider the following statements

(a) The system of has unique solution if $k \neq 2$, $k \neq -2$.

[Ans: **B**]

(b) The system has unique solution if $k = -2$

(c) The system has unique solution if $k = 2$

(d) The system has no solution if $k = 2$

(e) The system has infinite number of solutions if $k \neq -2$

Which of the following statements are correct?

[JEE Mains-2021]

(A) (a) and (e) only

(B) (a) and (d) only

(C) (b) and (e) only

(D) (c) and (d) only

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians